LITTORAL MATHEMATICS TEACHERS ASSOCIATION (L.M.T.A)

GENERAL CERTIFICATE OF EDUCATION REGIONAL MOCK EXAMINATION

0765 PURE MATHS WITH MECHANICS 2

MARCH 2023 ADVANCED LEVEL

Subject Title	Pure Mathematics With Mechanics
Paper No	2
Subject Code No	0765

THREE HOURS

Full marks may be obtained for answers to ALL questions.

Mathematical Formulae Booklets published by the Board are allowed.

In calculations, you are advised to show all the steps in your working giving the answer at each stage.

Calculators are allowed.

Start each question on a fresh page.

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1. Given that $p(x)$ is a polynomial of degree 3, and that $p(x)$ leaves a remainder of 12 and -8 when divided by $x-2$ and $x+3$ respectively.

(a) Find the remainder when $p(x)$ is divided by x^2+x-6 **4 marks**

Given also that $x-1$ is a factor of $p(x)$ and $p(0)=-2$

(b) Find the quotient when $p(x)$ is divided by x^2+x-6 **4 marks**

(c) Express $p(x)$ in the form ax^3+bx^2+cx+d where a, b, c, and d are constants. **2 marks**

2.

(i) Given that the roots of the quadratic equation $2x^2-5x+k=0$ are $(1+\alpha)$ and $(1+\beta)$, form a quadratic equation in terms of k whose roots are $(1+\alpha^2)$ and $(1+\beta^2)$ **5 marks**

(ii) The function $f: \Re/_{\left\{\frac{1}{2}\right\}} \to \Re/_{\left\{\frac{1}{2}\right\}}$ is defined

$$f: x \to \frac{1+x}{2x-1},$$

Show that f is bijective. **4 marks**

3.

(i) Find $\frac{dy}{dx}$ if:

(a) $y=e^{-2x}\ln(x^2-2)$ **3 marks**

(b) $x=2\cos 2\theta$, $y=2(\theta+\sin\theta)$ in terms of the parameter θ **3 marks**

(ii) Given the $f(x)=x^3-3x+7$ find the intervals in which the curve of $y=f(x)$ is

(a) strictly increasing. **2 marks**

(b) strictly decreasing. **1 mark**

4.

(i) Express $g(x)=\frac{4x+3}{(2x+1)(x+1)}$ into partial fractions.

Hence, find the first three terms in the binomial expansion of $g(x)$ in ascending powers of x. **6 marks**

(ii) Sketch the curve of $y=\frac{x-1}{x+3}$ showing clearly the points where the curve crosses the coordinate axes and the behaviour of the curve as it approaches its asymptotes. **5 marks**

5.

(i) The complex number z is such that

$$\frac{z+1}{1+i}+\frac{z}{1-i}=5+5i$$

Express z in the form $a+bi$ where a and b are real constants. **4 marks**

(ii) The relation R on the set of integers is defined by

$$xRy \Leftrightarrow (x-y) \text{ is divisible by } 3.$$

Show that R is an equivalence relation. **5 marks**

6.

(i) Evaluate the following:

(a) $\int_2^4 x^2 \ln(2x)dx$ **5 marks.**

(b) $\int \sin 5x \cos x dx$. **3 marks**

(ii) Find, in the form $y = f(x)$, the general solution of the differential equation

$$2(x+1)\frac{dy}{dx} = (2y+1). \quad \textbf{3marks}$$

7.

(i) Given the matrix $\mathbf{M}$, where

$$\mathbf{M} = \begin{pmatrix} 1 & -2 & 2 \\ -1 & 1 & -2 \\ 2 & -1 & 0 \end{pmatrix}$$

Find $\mathbf{M}^{-1}$ the inverse of $\mathbf{M}$.
Hence or otherwise, solve the system of equations

$$\begin{aligned} x - 2y + 2z &= 5 \\ -x + y - 2z &= -6 \\ 2x - y &= 5 \end{aligned}$$ **7 marks**

(ii) Solve for x in the equation $\log_e x + 2\log_x e = 3$. **4 marks**

8.

(i) Find the values of R and x for which $\sqrt{3}\sin\theta - \cos\theta = R\sin(\theta - \lambda)$, where $R > 0$ and $0 < \lambda < 90^0$.
Hence find

(a) the minimum value of $\dfrac{8}{6 + \left|\sqrt{3}\sin\theta - \cos\theta\right|}$ **5 marks**

(b) the general solution of the equation $\sqrt{3}\sin\theta - \cos\theta = -2$ **3 marks**

(ii) Proof by contradiction that $\sqrt{11}$ is irrational **4 marks**

9. The points A, B and C have coordinates $(0,1,3), (-1,2,-2)$ and $(-3,4,0)$ respectively.
Find

(a) $\mathrm{BA} \times \mathrm{BC}$ **3 marks**

(b) Cartesian equation the plane containing the points A, B and C. **3 marks**

(c) the sine of the angle $A\hat{B}C$ **3 marks**

10.

(i) Show that a root of the equation

$$2\ln x - \frac{1}{x} = 0$$

exists in the interval $1 < x < 2$.
Taking 1.5 as the first approximation to the root of this equation, use Newton – Raphson's method to obtain, to 2 decimal places , a second approximation to the root of this equation. **5 marks**

(ii) The sum of the first n terms of a progression is given by

$$S_n = 18\left(1 - \frac{1}{3^n}\right)$$

(a) Find the nth term of the progression . **2 marks**
(b) Show that it is a geometric progression stating the first term and common ratio. **3 marks**

END!

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